

DOI 10.24425/aee.2025.155959

Effect of different types of damping winding structures on the no-load magnetic field of a tubular hydro-generator

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(Received: 19.04.2025, revised: 03.11.2025)

Abstract: The no-load magnetic field of a hydro-generator significantly impacts the quality of its no-load voltage waveform and the grid power quality and power system stability. As a vital element for ensuring the safe and steady operation of hydro-generators, the damping winding structure directly affects the state of the no-load magnetic field. Particularly, horizontal hydro-generators, such as tubular turbine units, feature confined and irregular internal spaces that lead to more intricate and intense distributions of the magnetic field. Therefore, to improve the quality of no-load voltage waveforms, grid power quality, and overall power system stability, it is essential to examine how variations in damping winding structure types affect the no-load magnetic field in these generators. This paper considers a specific 34-MW large tubular turbine generator as an example. A 2D transient electromagnetic field model was developed to investigate the effects of four damping winding structures – fully damped, semi-damped, isolated damping bar, and solid-steel pole – on the magnitude and distribution of the no-load magnetic field, the quality of the no-load voltage waveforms, and the eddy-current losses within the damping system. The research directly supports the design and manufacturing processes of tubular hydro-generators and ensures the safety and stability of generator and power system operations.

Key words: damping winding structure, no-load magnetic field, power quality, tubular hydro-generator



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1. Introduction

The no-load magnetic field in a hydro-generator significantly influences the quality of the no-load voltage waveform and has implications for grid power quality and power system stability. The type of damping winding structure is a critical factor that impacts the generator's no-load magnetic field state, ensuring safe and stable operation. In horizontal tubular hydro-generators, designed to satisfy performance standards such as low head, high flow capacity, and hydraulic efficiency, the challenge lies in effectively arranging turbine and generator components within a constrained internal space. This compact configuration, depicted in Fig. 1, diverges from traditional vertical designs and results in complex distributions and dynamics of the no-load magnetic field within the limited, irregular internal space. Thus, it is crucial to investigate how different types of damping winding structures affect the no-load magnetic field in these generators. This research aims to optimize the magnetic field configuration to ensure the generation of high-quality no-load voltage waveforms, which will subsequently enhance grid power quality and strengthen power system stability.

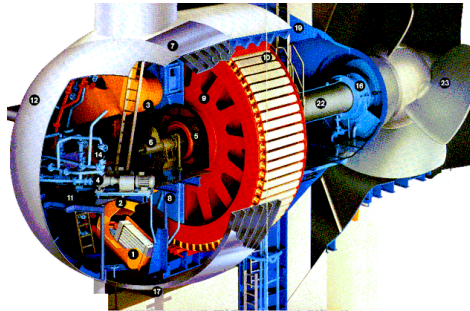


Fig. 1. Tubular hydro-generator structure

Hydro-generator damping winding structures can generally be categorized into four types, as outlined in Table 1. The two most common configurations are the fully damped and semi-damped structures [1, 2]. In the fully damped structure, all magnetic pole dampers are interconnected via end rings, forming a short-circuit arrangement where the damping windings of each pole are linked together, creating a cage-like formation, as illustrated in Fig. 2(a). Conversely, the semi-damped structure does not connect the damping windings between the poles, allowing each magnetic

Table 1. Damping winding structure schemes

Scheme	Structural description
1	Fully damped winding structure
2	Semi-damped winding structure
3	Isolated damping bar structure
4	Solid-steel pole structure

pole's damping winding to function independently and be self-contained, as depicted in Fig. 2(b). Additionally, there are two less common configurations: the isolated damping bar arrangement, which features damping bars that are not interconnected, as shown in Fig. 2(c), and the solid-steel pole arrangement that employs monolithic steel poles without dedicated damping windings, as illustrated in Fig. 2(d).

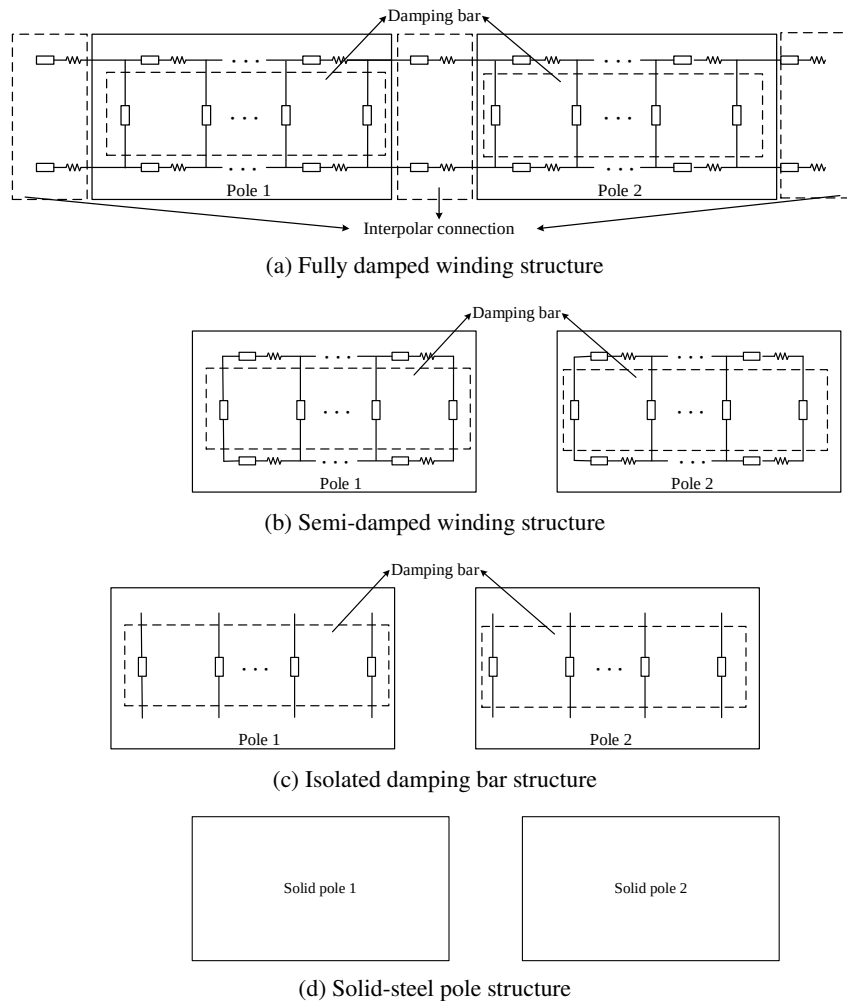


Fig. 2. Schematic diagram of the design scheme of the damping winding structure

Although substantial research has been conducted on the no-load magnetic field and voltage waveforms of hydro-generators [3–22], the majority of existing studies focus on specific localized structural parameters, such as the skewed stator slot [11–15], the offset of the magnetic pole boot center, the damping bar center offset [16–19], and the damping winding pitch size [20–22] regarding their impact on these characteristics. In contrast, there is a paucity of literature that

systematically examines the effects of fundamental damping winding structure types (e.g., fully damped, semi-damped, isolated damping bar, and solid-steel poles) on the no-load magnetic field and voltage waveform quality of tubular hydro-generators.

This paper examines a specific 34 MW tubular hydro-generator. By creating a 2D transient electromagnetic field finite element analysis model, the study explores how four typical damping winding structures – fully damped, semi-damped, isolated damping, and solid-steel pole – affect the magnitude and distribution of the no-load magnetic field, the quality of the no-load voltage waveform, and the eddy-current losses in the damping system. The findings provide direct support for improving the design and manufacturing standards of cross-flow hydro-generators, thereby contributing to the stability of power systems.

2. Modeling of tubular hydro-generator

2.1. Generator basic parameters

This research utilizes a 2D model of a specific 34 MW tubular hydro-generator operating under no-load conditions. Key parameters for the model are shown in Table 2.

Table 2. Key parameters of the tubular hydro-generator

Parameter	Value
Rated power (MW)	34
Rated voltage (kV)	10.5
Power factor	0.95
Number of phases	3
Number of poles	44
Number of slots per phase per pole	2
No-load excitation current (A)	547
Number of damping bars per pole	6
Stator inner diameter (mm)	5 620
Stator outer diameter (mm)	6 020
Rotor inner diameter (mm)	4 690
Rotor outer diameter (mm)	5 610
Minimal air gap (mm)	10
Maximum-to-minimum air gap ratio	≈ 1.5
Pole shoe width (mm)	265
Pole body width (mm)	175
Stator slot width (mm)	28
Number of turns of excitation winding	18